Euclidean Geometry In Mathematical Olympiads

Problem 1.50: IMO 2013/4

motivating us to show $HPWR$ is cyclic as $\angle WPH = 90^\circ = \angle WBX$.

Note: Because the first result is fully rigorous and mathematically correct, it's not a true *soft* technique, but it's unnecessary once you prove X, H, R are collinear.

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Aug 12, 2021, 12:59 PM *#11*

Solution

Denote the Miquel Point by P. Then P lies on (AMN), and since AMN is the orthic triangle the diameter is AH. This means $\angle APH = 90$. Since WX, WY are diameters, we have $\angle XPW = 90$ and $\angle YPW = 90$. We also have $\angle APY = 180 - \angle YPW = 90$, so points X, H, P and Y, H, P are collinear. This means points X, Y, H are collinear, as desired.

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Oct 15, 2021, 1:05 PM *#12*